

Silicon Photonics Layout Design Report

EBeam_NarehAsadorian.gds

Designer: Nareh Asadorian

Technology: SiEPIC EBeam PDK

Target polarization/wavelength: TE, 1550 nm

Design summary. The submitted layout contains four Mach-Zehnder-interferometer-based photonic circuits: two asymmetric MZI variants, a cascaded/double MZI, and a looped MZI. The circuits use grating couplers for optical I/O, 500 nm strip waveguides, Y-branch and broadband directional-coupler components, and a 1550 nm optical terminator. The final GDS contains SiEPIC verification metadata reporting zero layout errors.

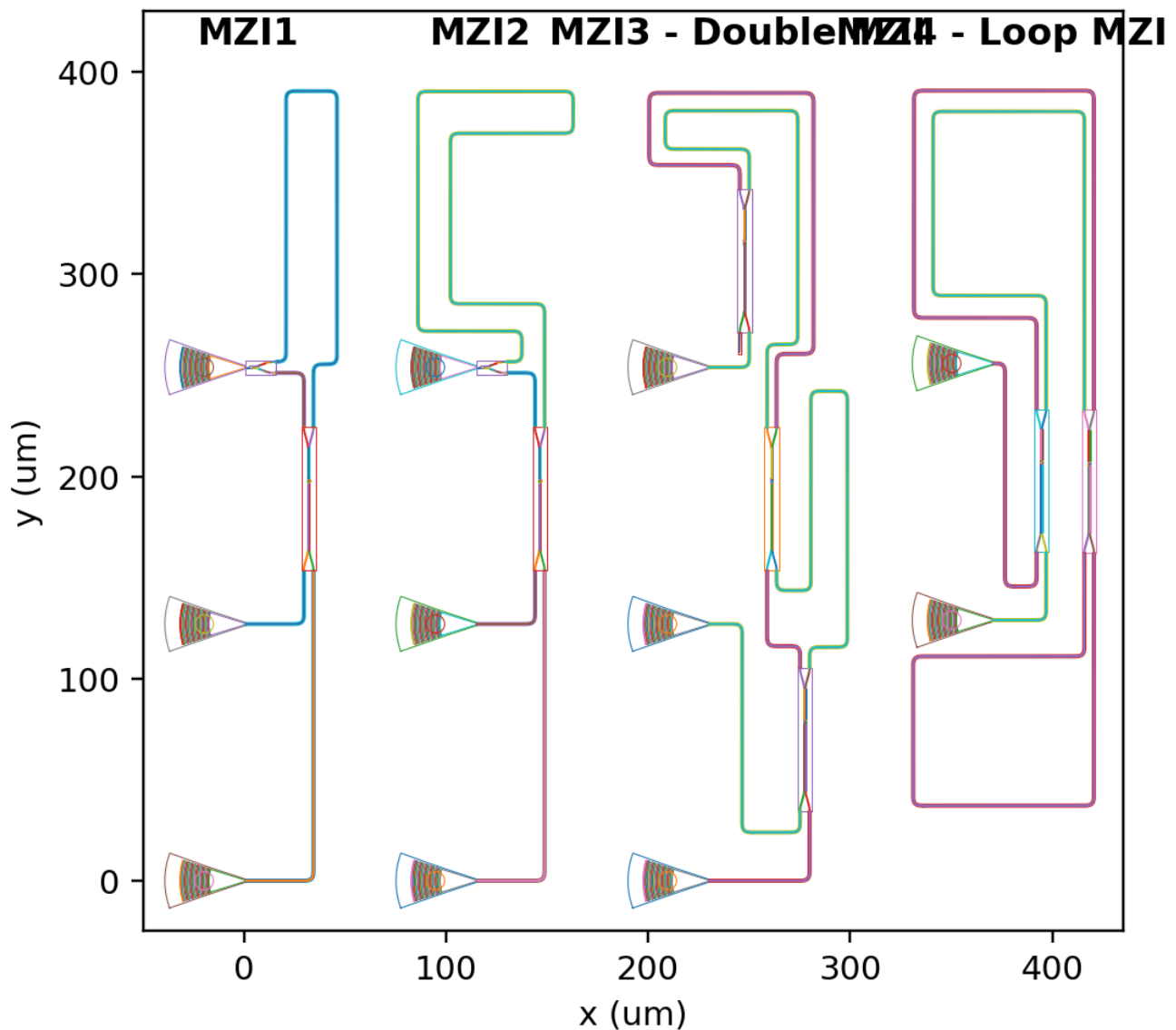


Figure 1. Overview of the submitted GDS layout reconstructed from the file hierarchy.

1. Design Objective

The objective of this layout is to implement and compare several interferometric silicon-photonics circuits suitable for automated optical characterization near 1550 nm. The designs intentionally vary optical path geometry and coupling topology while keeping a common measurement interface. This makes it possible to compare spectral interference, insertion loss, free spectral range, and the effect of increased path length or cascaded interferometers after fabrication.

2. Design Environment and Common Parameters

Parameter	Value / component
Layout tool	KLayout v0.30.12
SiEPIC-Tools	v0.5.31
Technology	EBeam PDK v0.4.53
Waveguide type	Strip waveguide, TE 1550 nm
Waveguide width	500 nm
Nominal bend radius	5 μm
Optical I/O	TE 1550 nm grating couplers
Splitters/couplers	Y-branch and 50/50 broadband directional couplers
Unused-port treatment	1550 nm optical terminator included in the final top-level layout

The grating-coupler cell in the GDS is identified as **ebeam_gc_te1550**, the Y-branch as **ebeam_y_1550**, the broadband 50/50 directional coupler as **ebeam_bdc_te1550**, and the termination structure as **ebeam_terminator_te1550**.

3. Circuit Designs

3.1 MZI1 - Asymmetric MZI

MZI1 uses one Y-branch, one 50/50 broadband directional coupler, three grating couplers, and several routed waveguide sections. The Y-branch divides the optical input into two arms. The unequal routed lengths create a relative phase difference between the arms, and the directional coupler recombines the fields. The resulting wavelength-dependent constructive and destructive interference is observed at the output ports. The longest dedicated routed section in this design is approximately 334.692 μm .

Expected behavior: A periodic transmission spectrum is expected. The spectral period is set primarily by the optical path-length difference between the two interferometer arms, while the extinction ratio depends on phase balance, splitting ratio, propagation loss, and fabrication variation.

3.2 MZI2 - Increased-Path-Length MZI

MZI2 keeps the same general Y-branch/directional-coupler architecture as MZI1 but replaces the long routed section with a substantially longer meandered path. The longest dedicated routed section is approximately 529.305 μm . This increases the arm imbalance relative to MZI1 and is intended to produce a stronger wavelength-dependent phase accumulation and a smaller free spectral range.

Design comparison: Because MZI1 and MZI2 share the same basic components and measurement interface, comparing their measured spectra provides a direct way to study how increased optical path difference changes the interference period.

3.3 MZI3 - Double / Cascaded MZI

MZI3 is a cascaded interferometric circuit containing three broadband directional couplers and three grating couplers. Multiple routed sections connect the couplers, with individual waveguide lengths ranging from approximately 35.938 μm to 348.218 μm . The topology forms two coupled interferometric stages rather than a single MZI. This enables a more complex transfer function than the first two circuits and can produce additional spectral structure through the superposition of phase responses from both stages.

Unused-port control: A dedicated 1550 nm terminator is included in the final top-level layout to terminate an otherwise unused optical port and reduce unwanted back-reflection into nearby waveguides.

3.4 MZI4 - Looped MZI

MZI4 uses two broadband directional couplers connected by long looped waveguide routes and two grating couplers. Two major routed sections are nearly equal in length (approximately 418.213 μm and 419.611 μm), while another long section is approximately 459.103 μm . The looped geometry creates an alternative interferometric path arrangement while keeping a compact footprint. It is useful for investigating phase accumulation in long routed arms and the spectral response of a loop-based MZI topology.

Table 2 summarizes the four circuit cells contained in the final top-level design.

Circuit	Main passive components	Grating couplers	Primary design purpose
MZI1	1 Y-branch + 1 directional coupler	3	Asymmetric MZI; long section $\sim 334.692 \mu\text{m}$
MZI2	1 Y-branch + 1 directional coupler	3	Larger path imbalance; long section $\sim 529.305 \mu\text{m}$
MZI3 (Double_MZI)	3 directional couplers	3	Two-stage/cascaded interferometer
MZI4 (MZI_Loop)	2 directional couplers	2	Looped interferometer; long arms $\sim 418\text{-}459 \mu\text{m}$

4. Measurement Interface and Labels

Each device is identified by an optical-input label compatible with SiEPIC automated measurement workflows: **opt_in_TE_1550_device_NarehAsadorian_MZI1**, ... **_MZI2**, ... **_MZI3**, and ... **_MZI4**. These labels identify the target polarization and nominal wavelength and associate the measured response with the corresponding circuit.

Grating couplers provide fibre-to-chip coupling for automated spectral measurements. The common 1550 nm TE interface allows all four circuits to be characterized with the same optical test setup.

5. Layout Verification and Fabrication Readiness

The final top-level cell is **EBeam_NarehAsadorian_v1**. Embedded SiEPIC verification metadata in the submitted GDS reports:

- SiEPIC-Tools verification: 0 errors
- Verification time recorded in the GDS: 2026-10-05 23:52:40
- SiEPIC-Tools v0.5.31

- EBeam technology v0.4.53
- KLayout v0.30.12

The top-level layout footprint extracted from the GDS is approximately 605 μm x 410 μm . The use of standard PDK components, explicit measurement labels, and a dedicated terminator improves compatibility with the fabrication and automated measurement workflow. Final fabrication acceptance still depends on the course repository checks and the foundry/PDK design rules.

6. Expected Optical Response

7. Simulation and Experimental Characterization Plan

8. Conclusion

The final layout implements four related silicon-photonics interferometer circuits on a common EBeam platform. The set progresses from conventional asymmetric MZI structures to a cascaded double MZI and a looped MZI, providing a compact test set for studying phase accumulation, path-length imbalance, coupling, and spectral interference near 1550 nm. The submitted GDS uses standard SiEPIC PDK components and contains embedded verification metadata reporting zero SiEPIC layout errors.

Appendix A - Extracted Waveguide Segment Lengths

Circuit	Waveguide segment lengths in GDS (μm)
MZI1	39.478, 53.878, 185.578, 334.692
MZI2	39.478, 53.878, 185.578, 529.305
MZI3	35.938, 61.527, 82.918, 154.295, 290.020, 308.610, 348.218
MZI4	58.358, 144.315, 418.213, 419.611, 459.103